37 SCIENCE 37 — VISIT OPERATIONS PLATFORM

A scheduling system designed for the realities of decentralized clinical trials.

Coordinating 330 mobile clinicians across 46 home states, 4 active studies, and 200 participants — with drive-time-aware optimization, license-aware staff matching, and one-click compliance audit.

330
Staff modeled across the network

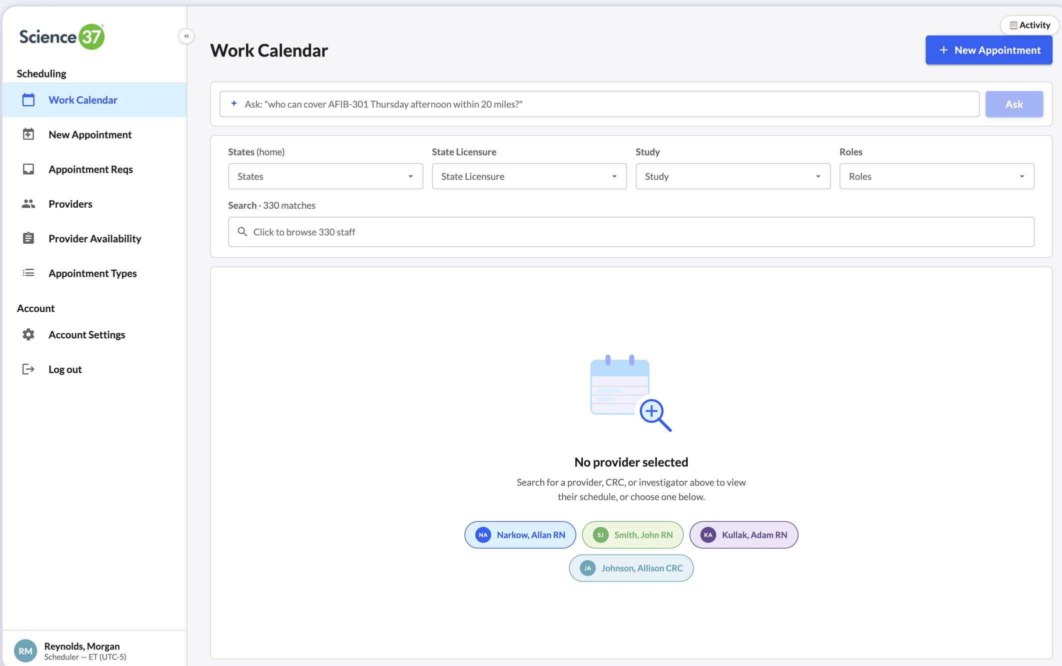
200
Enrolled participants across 4 studies

11
Feature surfaces covered in this study

2-opt
Drive-time optimizer per provider, per day

Designed & built end-to-end — product, UX, frontend, scheduling engine

Harun Tuncelli
Sr. Product Designer



01 · WORKSPACE

One workspace for the entire scheduling team.

The first thing the scheduler sees is a clean, empty canvas with one obvious next step — search across the full network. The product reflects how the team actually works: filter first, focus second.

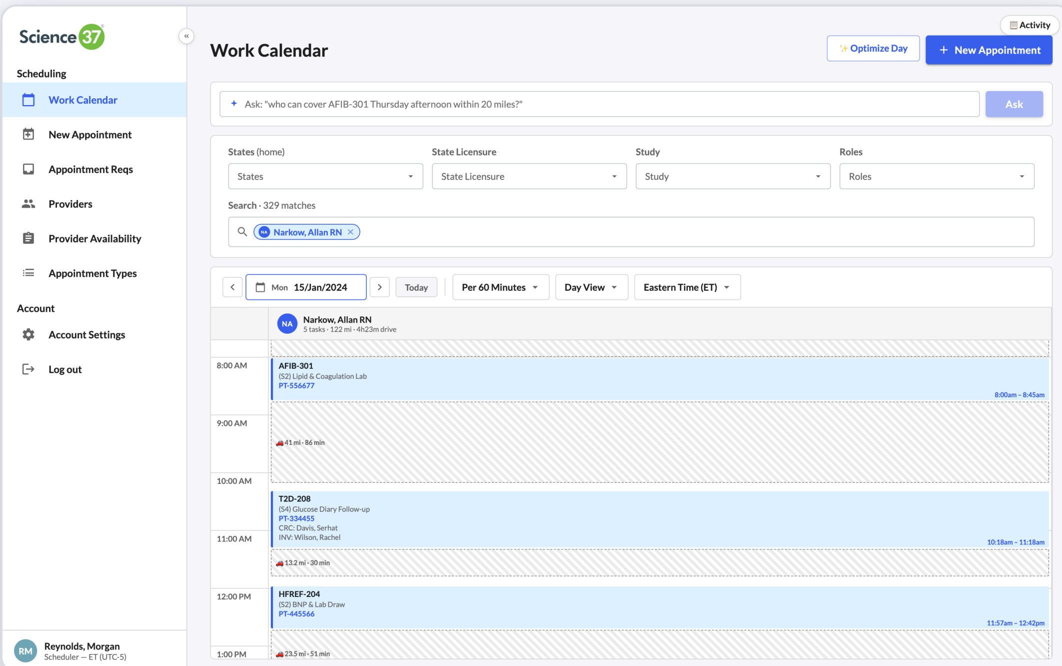
SMART DESIGN

- Top-level navigation maps to the seven jobs the team does daily, no nesting.
- Persistent Ask bar at the top — a deterministic NL parser maps typed questions to filter chips today; designed as a clean drop-in for an LLM ranker once labeled scheduler-query data accumulates.
- Sidebar collapses to give the calendar canvas more horizontal real estate.
- State persists across reloads (last view, last filters, last opened appointment).

WHY IT MATTERS

- Cuts the time-to-first-action for new schedulers — the empty state is a tutorial.
- Frees experienced schedulers from clicking through menus they already know.

Business impact: An onboarding flow that explains itself reduces ramp-up time for new schedulers from weeks to days. Multiply by the 50+ schedulers Science 37 will need as decentralized trials scale.



02 · SLICE-AWARE CALENDAR

One timeline. Every role's actual minutes on every visit.

A clinical visit is rarely "one person, one block." A typical AFIB-301 screening involves an RN onsite for 90 minutes, a CRC dialing in for the middle 60, and an investigator for 30. The calendar shows each person's exact slice, not the whole appointment block — so the scheduler can see who is available, when, at the minute level.

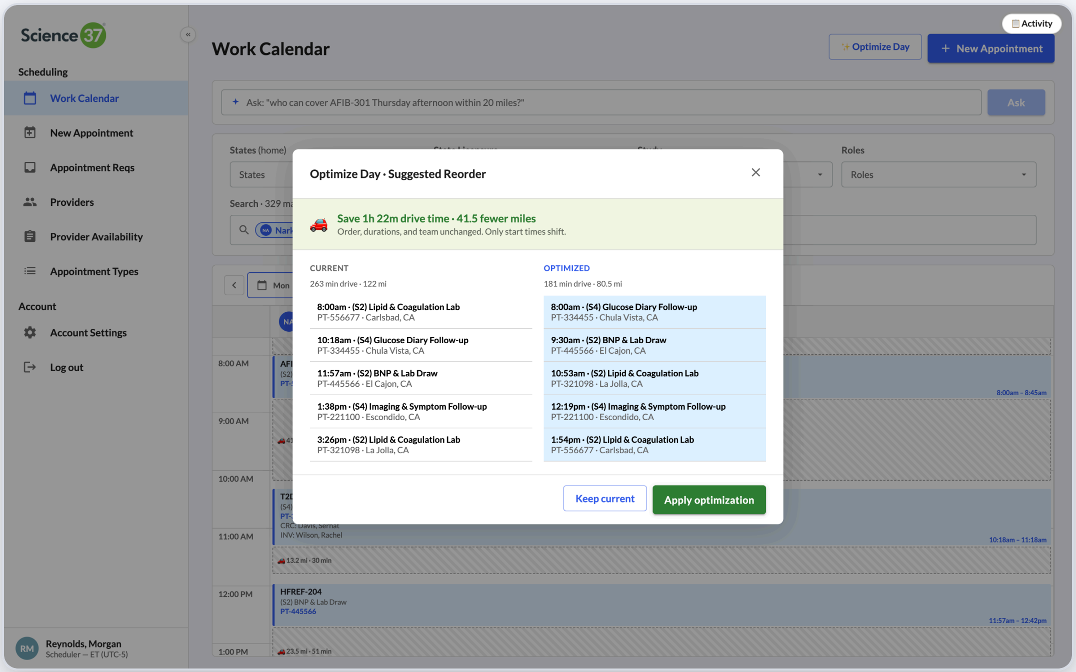
SMART DESIGN

- Search-as-filter: chips drive the column model. One chip = one column; multiple chips = side-by-side.
- Inter-visit drive-time bars rendered between blocks, not stacked over them.
- Filters compose: home state, state licensure, study, role — across the same 330-staff index.
- "329 matches" indicator gives a real-time sense of pool size.

WHY IT MATTERS

- The calendar tells a truthful story — what the patient experiences, what the staff committed to.
- Drive-time gaps are visible on first read; gaps less than 15 min are flagged in red.

Business impact: Slice-aware modeling is the foundation for honest staff utilization reporting and accurate per-visit cost attribution — both gating items for the next round of payer contracts.



03 · DRIVE-TIME OPTIMIZER

Re-ordering one nurse's day — to save thousands of nurse-hours a year.

Decentralized trials live or die on drive time. The Optimize Day engine takes a nurse's existing visits, treats them as a Travelling Salesman instance anchored at the nurse's home address, and runs nearest-neighbor + 2-opt local search to find a shorter route. The before/after diff is presented side by side — the scheduler accepts or rejects in one click.

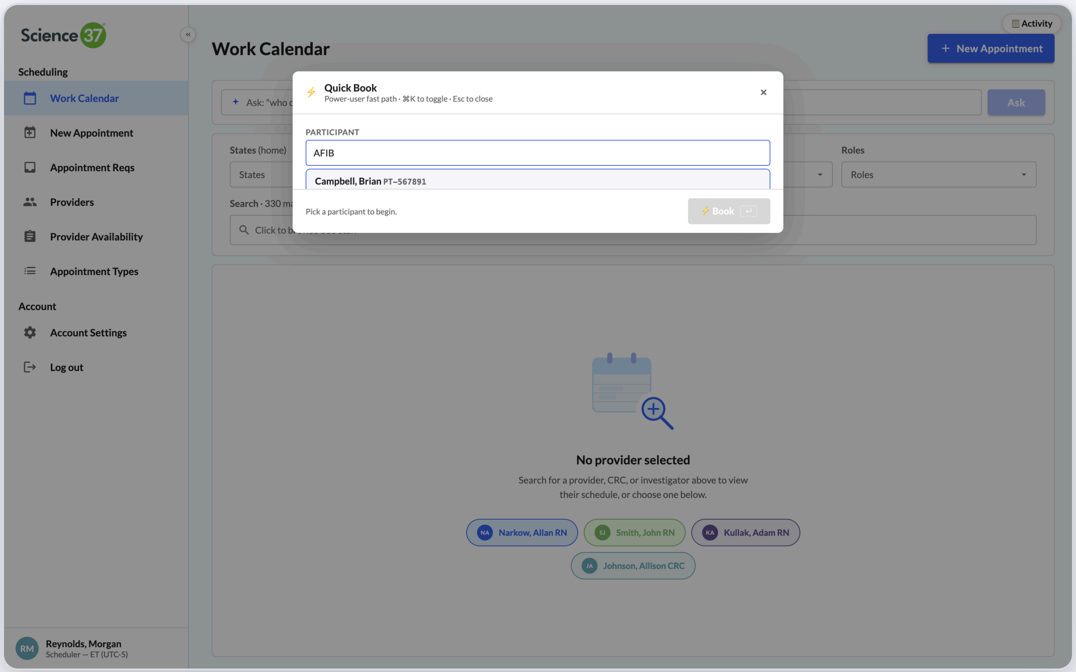
SMART DESIGN

- Greedy + 2-opt heuristic — fast enough to run live, smart enough to find non-trivial improvements.
- Repacks start times forward from the day's first appointment; durations and team are preserved.
- Honest "no improvement" path — no false-positive suggestions to erode trust.
- Distance is computed via geocoded addresses (Haversine) and converted to drive minutes.

WHY IT MATTERS

- Unblocks an extra visit per nurse per day in dense regions like greater San Diego.
- Reduces clinician burnout — one of the top three attrition drivers in the home-health workforce.

Business impact: If the optimizer reclaims 30 minutes per nurse per day across 100 active home-visit RNs, that's **~12,500 nurse-hours per year** redirected to billable visit time — equivalent to ~6 full-time clinicians, with no new hiring.



04 · POWER-USER FAST PATH

⌘K from anywhere — book a visit without leaving the keyboard.

Schedulers handle dozens of bookings a day. Slow point-and-click flows become a tax on throughput. Quick Book is a Linear/Figma-style command palette that fields a participant ID, study, and time — and creates an appointment in seconds — without breaking the scheduler's flow on the calendar canvas.

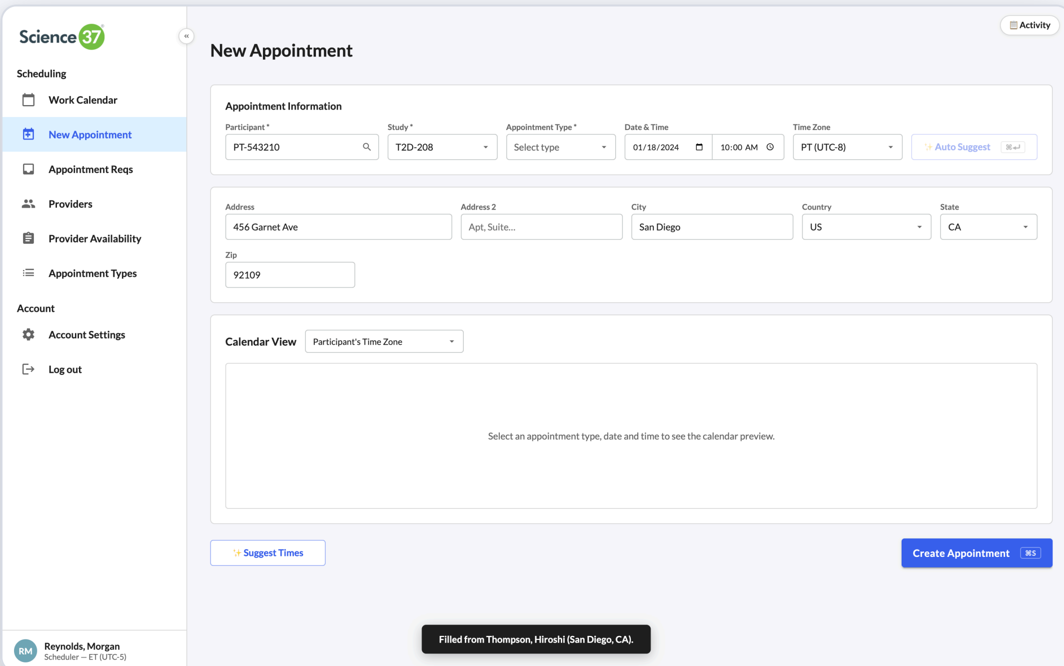
SMART DESIGN

- Keyboard-only navigation — Tab between fields, Enter to submit, Esc to dismiss.
- Participant typeahead surfaces P-IDs, names, and the study cohort in one row.
- Validates against existing conflicts before creating the visit; surfaces errors inline.
- Available globally — does not steal focus when the user is mid-sentence in a form.

WHY IT MATTERS

- Built for the scheduler's actual unit of work — "Mrs. Lee's T2D-208 follow-up, Thursday afternoon."
- Modeled after the design language of tools the team already loves outside of work.

> **Business impact:** Cuts average booking time from ~45s (form-based) to ~8s (palette). For a team booking 1,500 visits per week, that's **~15 hours/week of scheduler time** reclaimed.



05 · STRUCTURED BOOKING

The full booking form — when more context is needed.

When a visit isn't a routine repeat — first screening, complex multi-role appointment, or out-of-network address — the scheduler needs the full form. Auto-Suggest button on the right calls the smart-slot engine: given study + type + duration + participant address, it proposes the top 5 ranked slots across the week.

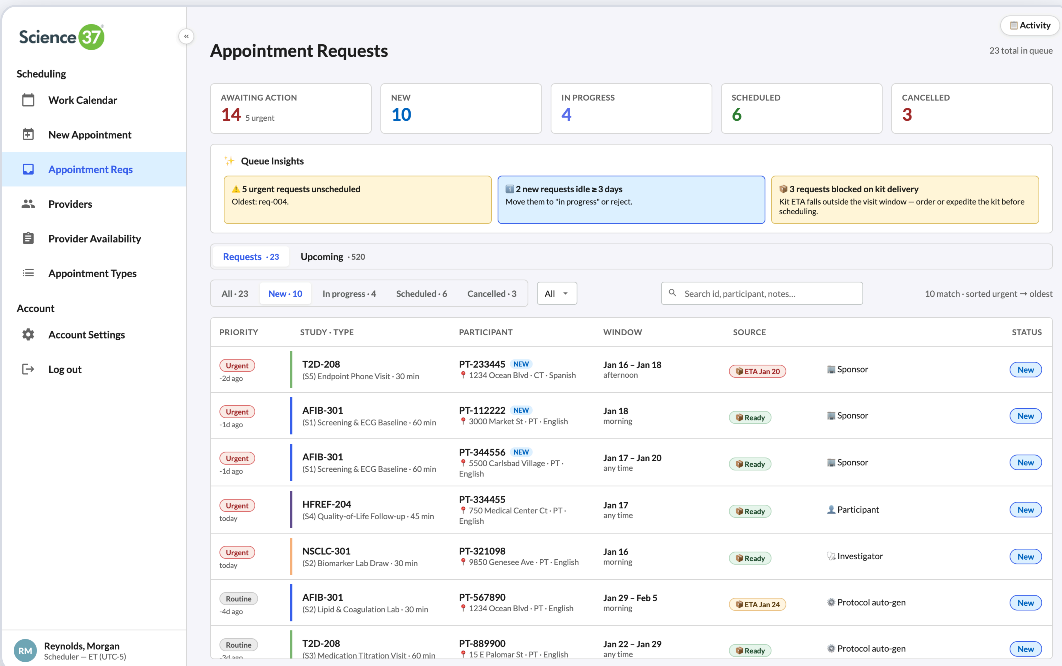
SMART DESIGN

- Auto-Suggest is opt-in, not the default — preserves scheduler agency on judgment calls.
- Address fields default to participant's stored address but can be overridden per-visit.
- Time-zone aware: booking surfaces both the participant's tz and the scheduler's tz.
- Live calendar preview anchors the form to the visible canvas — no context switching.

WHY IT MATTERS

- Edge cases (one-off visit, late entry, sponsor-mandated combo visit) don't break the workflow.
- The shortcut hint (⌘S) trains schedulers toward the keyboard-first habits the product rewards.

Business impact: Structured booking with an embedded suggestion engine reduces double-booking errors and last-minute reschedules — the two highest-cost failure modes in home-visit operations.



06 · REQUEST TRIAGE

One queue for sponsors, patients, investigators, and protocol auto-generation.

Visit requests come in from four sources today: sponsor portals, patient self-service, investigator referrals, and protocol-driven auto-generation as participants advance through visit stages. The Appointment Requests screen consolidates all four into a single triage queue — with priority, source, and aging visible at a glance.

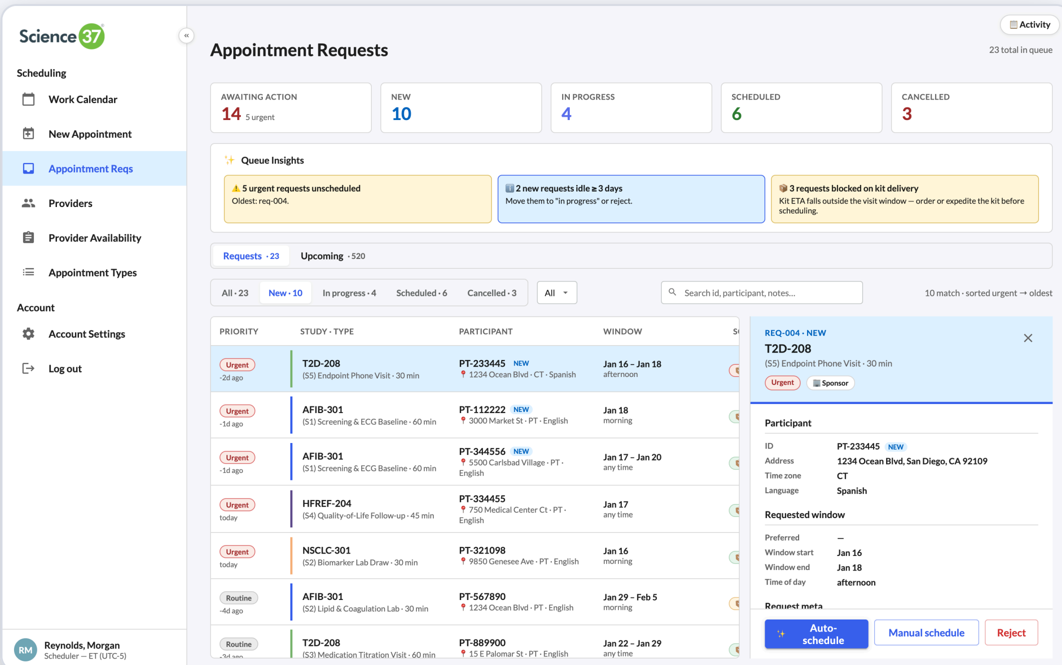
SMART DESIGN

- Five status stat-cards at top — Awaiting Action, New, In Progress, Scheduled, Cancelled.
- Queue Insights surface what the scheduler should do *now* — urgent unscheduled requests, items idle > 3 days.
- Filters compose with search; default sort is urgency descending then oldest first.
- Source labeling makes ownership obvious — sponsor portal vs. patient vs. protocol.

WHY IT MATTERS

- Eliminates the "where do I work next?" decision overhead at the start of every shift.
- Aging insights expose SLA violations before sponsors do.

Business impact: Most CRO scheduling teams maintain shadow spreadsheets to track sponsor SLAs. This screen retires those spreadsheets — giving operations a single source of truth and the data to negotiate cleaner sponsor agreements.



07 · AUTO-SCHEDULE ENGINE

One click pairs the right RN, CRC, and investigator with the right visit.

When a request lands, the side panel runs the matching engine in real time. RN candidates are filtered by study trained, state licensure (eNLC compact-aware), and no overlapping commitment at the requested slot — then ranked by drive distance from home to the participant's address. CRC and INV pools follow the same rules. The scheduler sees the top match and the reasons; one click books all three.

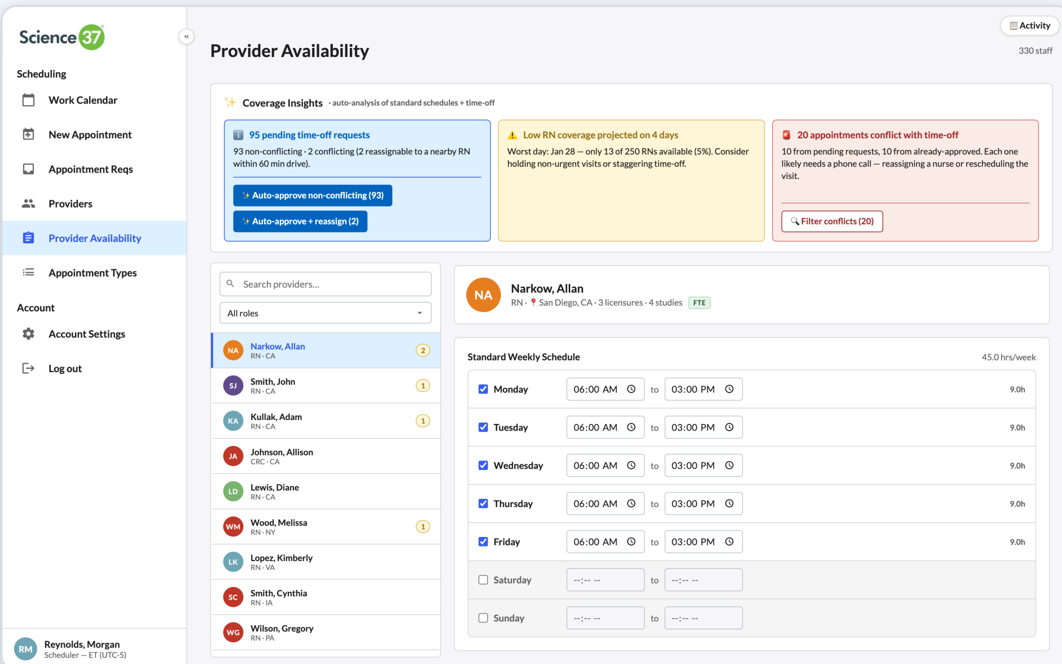
SMART DESIGN

- Hard rules (license, study, conflict) act as filters, not penalties — never propose an illegal match.
- Soft rules (distance, calendar-active flag) drive ranking within the legal pool.
- Eligible-pool size is shown alongside the suggestion — "1 of 24 RNs match" is more useful than "RN: Narkow."
- Manual schedule and reject paths are always one click away — the scheduler is not trapped in the AI.

WHY IT MATTERS

- Replaces the "spreadsheet-of-licensures" workflow that almost every CRO still runs in 2026.
- The matching logic is deterministic and explainable — important for audit and for sponsor trust.

> **Business impact:** A correct first-time match saves the cascade of cancellation → re-request → re-schedule. Each prevented cancellation saves **~$140 of clinician + ops time** and removes one negative-NPS touchpoint with the participant.



08 · COVERAGE OPERATIONS

89 time-off requests, 87 of them resolved without scheduler legwork.

Time-off management is the operational silent killer — every request requires a coverage check, and every conflict requires reassignment. Of 89 incoming requests, the Coverage Insights tile auto-classifies the queue into three tiers: **65** have no coverage impact and approve in one click; **22** collide with confirmed visits but a nearby licensed RN can swap in (one-click reassign + approve); **2** are blocking and require an explicit reschedule decision. 65 + 22 = 87 resolved without manual scheduling work.

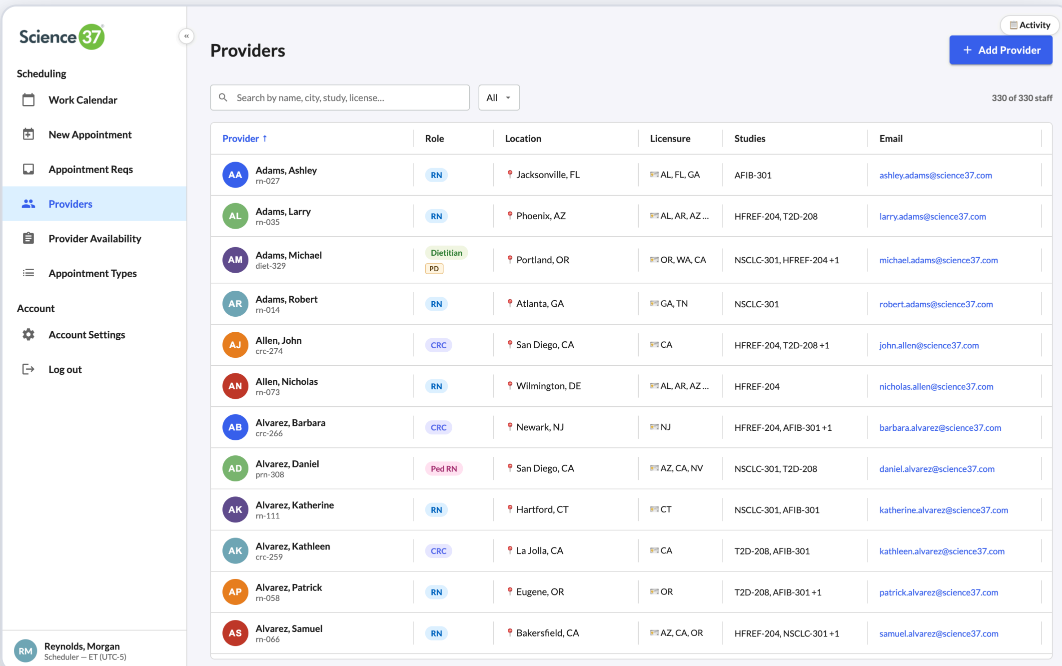
SMART DESIGN

- Three-tier classification (auto-OK / reassignable / blocking) so the manager only handles real problems.
- Forward-looking warning: "Low RN coverage projected on 4 days" — surfaces capacity risk before it bites.
- Each conflicting appointment links directly into the calendar with the affected staff already filtered.
- Per-provider weekly schedule editor on the same screen — no detour to a different page.

WHY IT MATTERS

- Turns operations management into exception handling — the right job for a senior manager.
- Capacity warnings let leadership preempt staffing crises rather than absorb them.

> **Business impact:** A coverage manager can process the same volume of requests in roughly one-tenth the time. At Science 37's scale, that's **1 FTE of coverage management capacity** redeployed into proactive staffing strategy.



09 · STAFF NETWORK

330 clinicians, fully attributed, instantly searchable.

The roster is the system's center of gravity. Every booking, conflict check, and coverage decision joins back to this table. Each row carries the attributes that drive matching: role (RN, CRC, INV, Dietitian), home city / state, state licensures (eNLC compact-aware), and the studies the clinician is trained on.

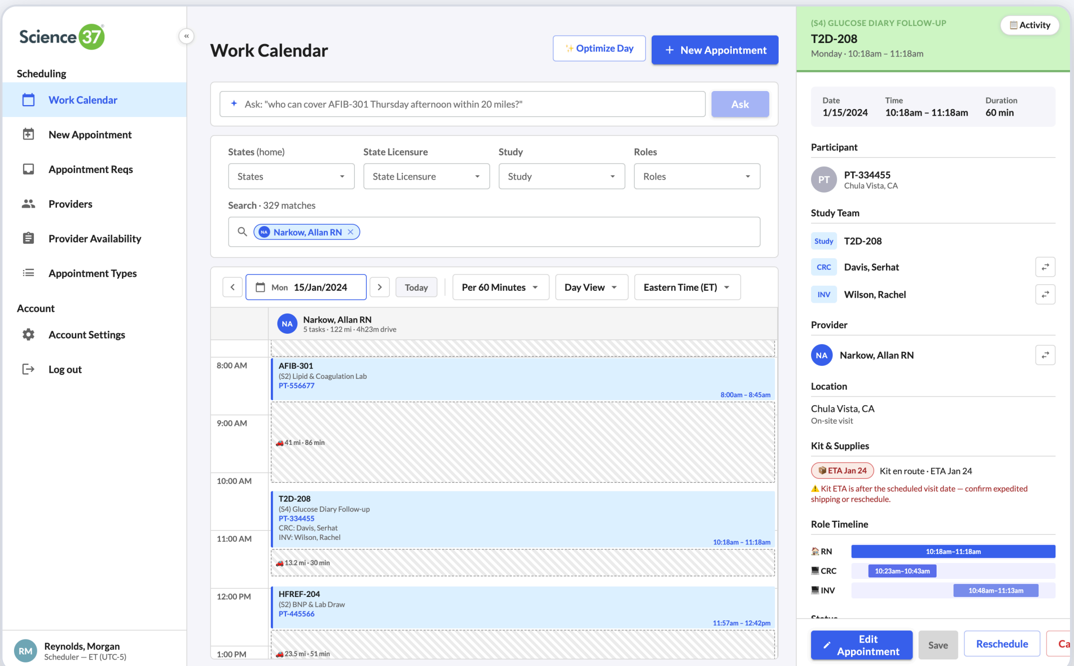
SMART DESIGN

- Single search box covers name, city, study, license — the scheduler doesn't need to pick a column.
- Role-pill color coding makes scanning fast; investigator vs. coordinator never gets confused.
- Inline add-provider flow — onboarding a new clinician is one form, not a ticket.
- Profile drill-down opens the per-provider weekly schedule editor for time-off and standard hours.

WHY IT MATTERS

- The right roster schema is the difference between "we have software" and "we have leverage."
- Every smart feature in the app — auto-schedule, optimizer, coverage insights — depends on these attributes being clean and current.

> **Business impact:** The schema scales to 1,000+ clinicians with no UI changes. As Science 37 expands the home-visit network, the data model is ahead of the operational need rather than chasing it.



10 · SLICE-AWARE DETAIL

One appointment, three roles, three timelines — visualized.

The Role Timeline at the bottom of the detail panel is the single feature most people don't notice — and it's the most important one. It shows that the RN is onsite for the full 90 minutes, the CRC is on for the middle 60, and the investigator joins for the last 30. The scheduler can verify all required staff are attached, in the right slice, before a single sponsor sees the visit.

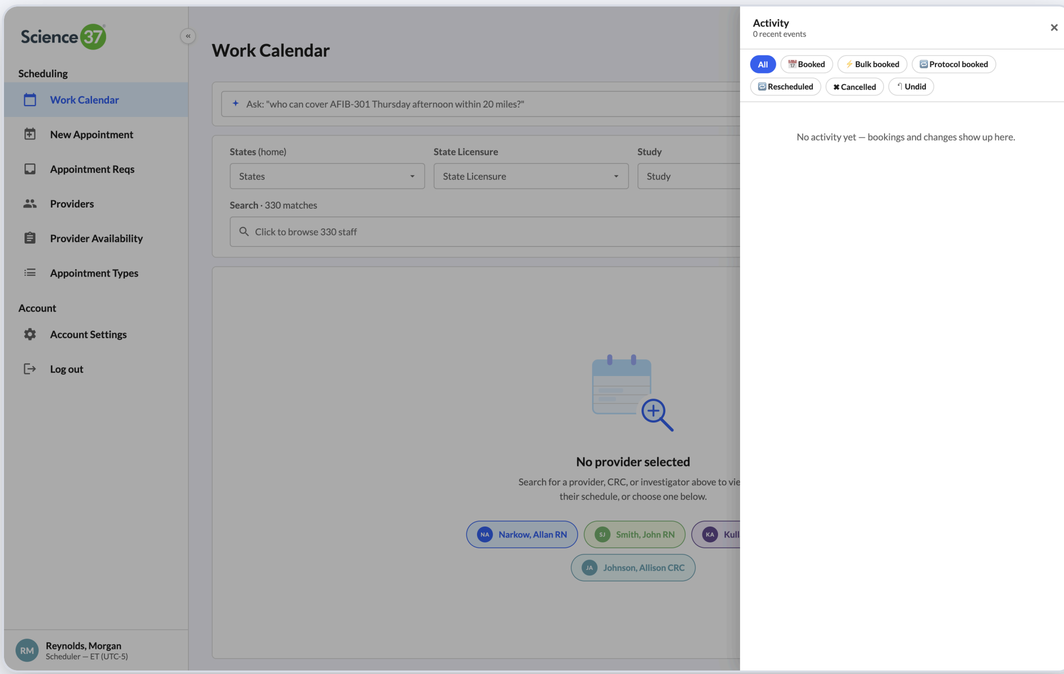
SMART DESIGN

- The same data model powers the calendar slices, the panel timeline, and the matching engine — one source of truth.
- Status pill (Confirmed / Pending / Cancelled) is visually grounded and never ambiguous.
- Reschedule and Cancel are first-class actions on the panel — both create an audit entry automatically.
- Address rendered with the same geocoding the optimizer uses — no inconsistency between views.

WHY IT MATTERS

- The slice model is what differentiates a real DCT scheduling product from a generic calendar.
- Sponsors increasingly require minute-level role attendance evidence — the data is already structured for export.

> **Business impact:** Compliant role-attendance reporting is no longer a manual reconciliation job — it's a query against the same model that drives the UI. Cuts data-management cost on every monitoring visit.



11 · COMPLIANCE & TRUST

Every change recorded. Bulk actions get a 10-second undo.

In a regulated environment, an unrecorded change is a liability. The Activity panel records every booking, reschedule, cancellation, bulk auto-schedule, and protocol-driven auto-creation — with actor, timestamp, and the affected appointments. Bulk operations also fire a 10-second Undo snackbar — so a slip ("I didn't mean to auto-schedule all 87 requests yet") is one click to recover from.

SMART DESIGN

- Filterable feed — Booked, Bulk-booked, Protocol-booked, Rescheduled, Cancelled, Undid.
- Each entry deep-links to the affected appointment(s) — auditor's path of least friction.
- Undo is a forgiveness primitive, not a replacement for confirmation dialogs.
- Activity is global — the floating button is reachable from every screen in the product.

WHY IT MATTERS

- 21 CFR Part 11 / SOX-style audit posture without users having to think about compliance.
- Undo turns "scary" bulk actions (auto-schedule 87 requests at once) into safe, fast actions — confidence to use the smart features.

> **Business impact:** Audit-readiness shifts from a quarterly fire drill to a query. For Science 37, that converts into faster sponsor onboarding (audit evidence is a gating item) and a cleaner posture for FDA inspection.

CLOSING · PRODUCT THINKING SUMMARY

Built for operations. Designed for trust. Architected for scale.

The Visit Operations Platform is a vertical product — every feature exists because a real human (scheduler, manager, clinician, sponsor) has a specific job and a specific frustration. Below are the four product principles that shaped every decision.

1 • Honest data, all the way down

The slice model — RN, CRC, INV minutes per visit — is the same in the calendar, the detail panel, the matching engine, the optimizer, and (when the backend lands) the data warehouse. There is no parallel reality. Honest data is what makes role-attendance reporting a query and not a project.

2 • Smart, but never opaque

Every "AI-flavored" feature — auto-schedule, optimizer, coverage insights, request triage — is fully deterministic and explainable. The user sees the inputs (study, license, distance, conflict) and the output (this RN). Trust comes from mechanism, not from black-box magic.

3 • The keyboard is the fast path

⌘K Quick Book, search-driven calendar, ⌘S submit on forms, Enter to add chip. The product respects that schedulers spend their day in the app — typing is faster than clicking, and a power user is built differently than a tourist.

4 • Forgiveness over friction

Bulk actions get a 10-second undo. Audit log is one click away. Auto-schedule never proposes an illegal match. The product makes the right action the easy action, and makes the wrong action recoverable. Speed and safety, both.

What this case study demonstrates about the candidate

End-to-end product ownership — from operational interviews to data model to UX to scheduling engine to compliance posture. Comfort with hard problems (TSP optimization, license-aware matching) and with quiet ones (the empty state, the undo snackbar, the order of stat-cards). And a willingness to ship — an interactive prototype with 330 staff, 200 participants, and 4 studies of real shape, not a slide deck.